

# SHIKHA VIRENDER CHANDEL

AI ENGINEER | DATA SCIENTIST | PhD RESEARCHER

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## PROFESSIONAL SUMMARY

AI/ML engineer and Ph.D. researcher building computer vision, LLMs/RAG, and data pipelines using PyTorch and Python. IEEE-published researcher with NASA- and NSF-funded work and production experience supporting 1M+ daily jobs.

## EDUCATION

**Ph.D. in Applied Science and Technology** — North Carolina A&T State University **Expected May 2028 | GPA 3.9**  
**M.S. in Computer Science** — University of North Carolina at Charlotte **May 2025 | GPA 3.8**

## TECHNICAL SKILLS

**Programming and Data:** Python, SQL, Java, Go, JavaScript, Pandas, NumPy, PostgreSQL, SQL Server  
**Machine Learning:** PyTorch, TensorFlow, scikit-learn, OpenCV, CNNs, EfficientNet, Vision Transformers, Segmentation, Classification, Visualization, Data Engineering, Data Modelling  
**Artificial Intelligence:** LLMs, RAG, OpenAI API, Vector Embeddings, Sentence Transformers, FAISS, TF-IDF, NLP  
**MLOps & Engineering:** FastAPI, Streamlit, Docker, REST APIs, Git(Version control), CI/CD, Linux, Redis, AWS

## WORK EXPERIENCE

**Graduate Research Assistant** **August 2025 - Present**  
North Carolina Agricultural and Technical State University Greensboro, NC

- Fine-tuned deep learning models for semantic segmentation and multi-label classification for Building Damage Assessment, benchmarking performance across 9,000+ Optical images using HuggingFace Transformers and Foundational Models.
- Researched and designed(R&D) an end-to-end ML experimentation pipeline spanning data preprocessing, model training, cross-validation, visualization, and ablation studies; evaluated models with F1-score, IoU, Dice, precision, and recall.

**Graduate Teaching Assistant** **August 2024 - May 2025**  
University of North Carolina - Charlotte Charlotte, NC

- Mentored master's and Ph.D. students in Java, Python, AI/ML algorithms, model development, debugging, and performance optimization across multiple course sections

**Software Developer** **June 2022 - April 2023**  
Cogo Port Private Limited Mumbai, India

- Reduced REST API response time by 50% through query optimization and Redis caching; optimized Sidekiq workflows processing 1M+ background jobs per day
- Built and deployed a Dockerized ticket-management microservice in Go, eliminating a third-party dependency; served as product/code owner and Scrum Master for reviews, sprint planning, and CI/CD integration.

**Machine Learning Intern** **July 2020 - September 2020**  
Verzeo Remote, India

- Developed an NLP sentiment-analysis pipeline supporting data-driven decisions for 12 analysts and improved deployment efficiency by 15% through CI/CD workflow enhancements.

## SELECTED PEER-REVIEWED IEEE PUBLICATIONS

**Bridging Spatial and Frequency Views for Disaster Assessment: Benefits and Limitations**  **June 2026**

**A Unified Framework for Efficient Remote Sensing Visual Question Answering: Adapting Dual, Hybrid, and Encoder-Decoder Architectures**  **June 2026**

**Comparison of Deep Learning Frameworks For Rice Disease Mapping From UAV Multispectral Imaging**  **June 2026**

## SELECTED AI PROJECTS

**Legal Information Retrieval Pipeline** /LLM-Agent-Legal-Retrieval **April 2026**

- Compare a hybrid sparse and dense retrieval pipeline with different LLM fine tuning over 269K+ legal records and evaluated classification performance using macro-F1 for a multilingual legal retrieval competition.

**AI-Powered Educational Assistant (RAG System)** /Streamlit /StudyBudy-AI **May 2025**

- Built and deployed an LLM learning assistant using embedding-based retrieval and query refinement, delivering an end-to-end workflow for transcript summarization, quizzes, and contextual Q&A